

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS

LECTURE-01

PHYSICS

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Topics To Be Covered

- 1** Fluids
- 2** Density
- 3** Pressure
- 4** Atmospheric Pressure
- 5** Variation Pressure with Depth

MECHANICAL PROPERTIES OF FLUIDS

Pressure due to a fluid column; Pascal's law and its applications (hydraulic lift and hydraulic brakes), effect of gravity on fluid pressure. Viscosity, Stokes' law, terminal velocity, streamline and turbulent flow, critical velocity, **Bernoulli's theorem** and its simple applications. Surface energy and surface tension, angle of contact, excess of pressure across a curved surface, application of surface tension ideas of drops, bubbles and **capillary rise**.

lecture no. 1

①

Fluids (liquid & gas)

- fluid are the substances that flow because they have low intermolecular forces (weak) and more intermolecular space

Liquid 9th

Gas 9th

- Interatomic distance medium
- Interatomic force medium

flow

no definite shape



- Interatomic distance large



- Inter-atomic force weak

flow

no definite shape

Density

$\rho \rightarrow$ Rho

It is mass per unit volume

$$\rho = \frac{m}{V} = \frac{gm}{cm^3} \quad (C.G.S) \quad \text{Neet}^2$$

$1cm^3 = 1ml$

Density = $\frac{\text{mass}}{\text{Volume}}$ # Note

$$\rho = \frac{m}{V}$$

(C.G.S)

Density of water $\rho_w = 1 \frac{gm}{cm^3} = 1 \frac{gm}{ml}$

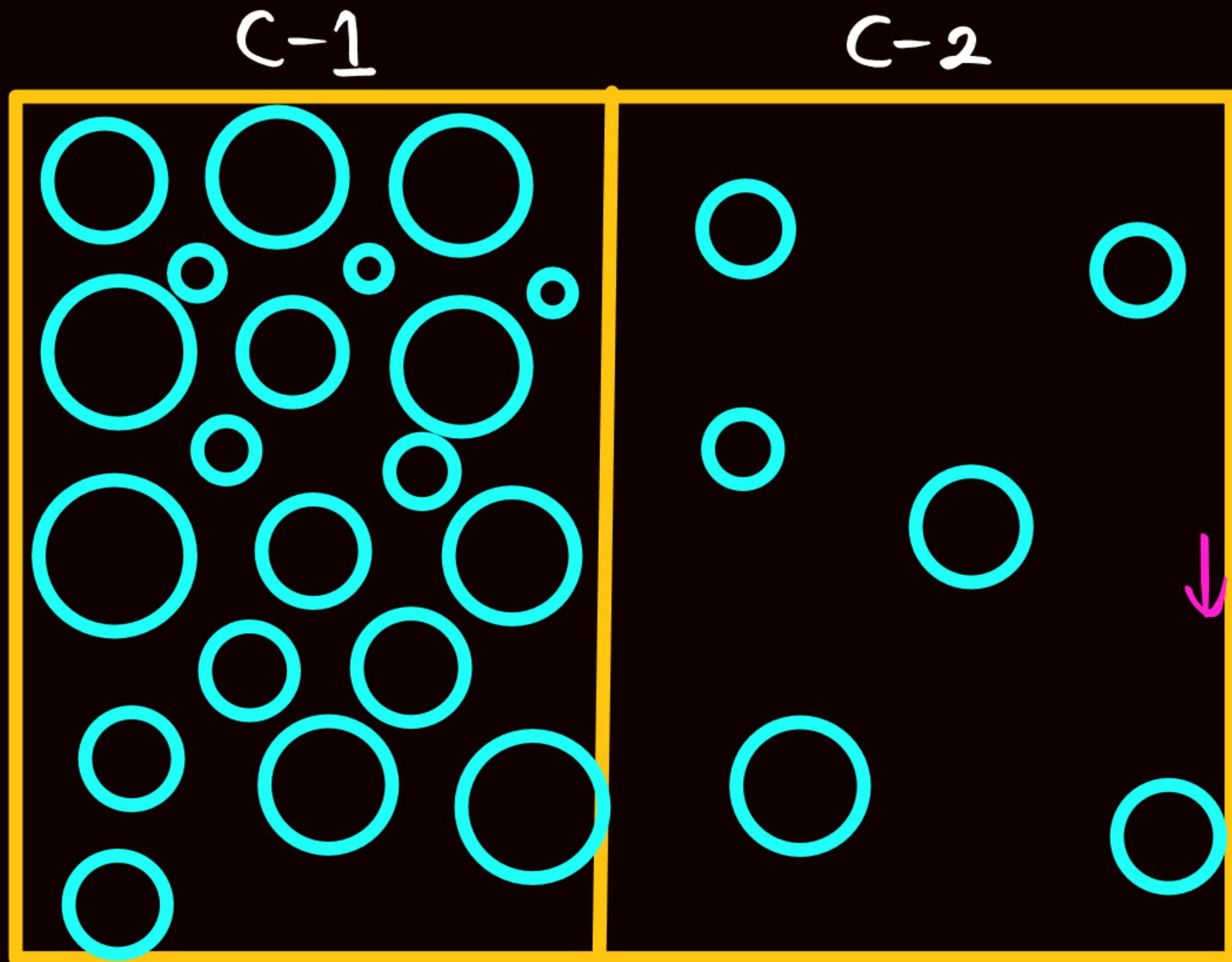
unit

$$\rho = \frac{m}{V} = \frac{kg}{m^3} \quad (S.I)$$

S.I Density of water $\rho_w = 1000 kg/m^3$



$$\uparrow \rho = \frac{m \uparrow}{V \downarrow}$$



C-1

C-2

3

Ⓐ C-1

Ⓑ C-2

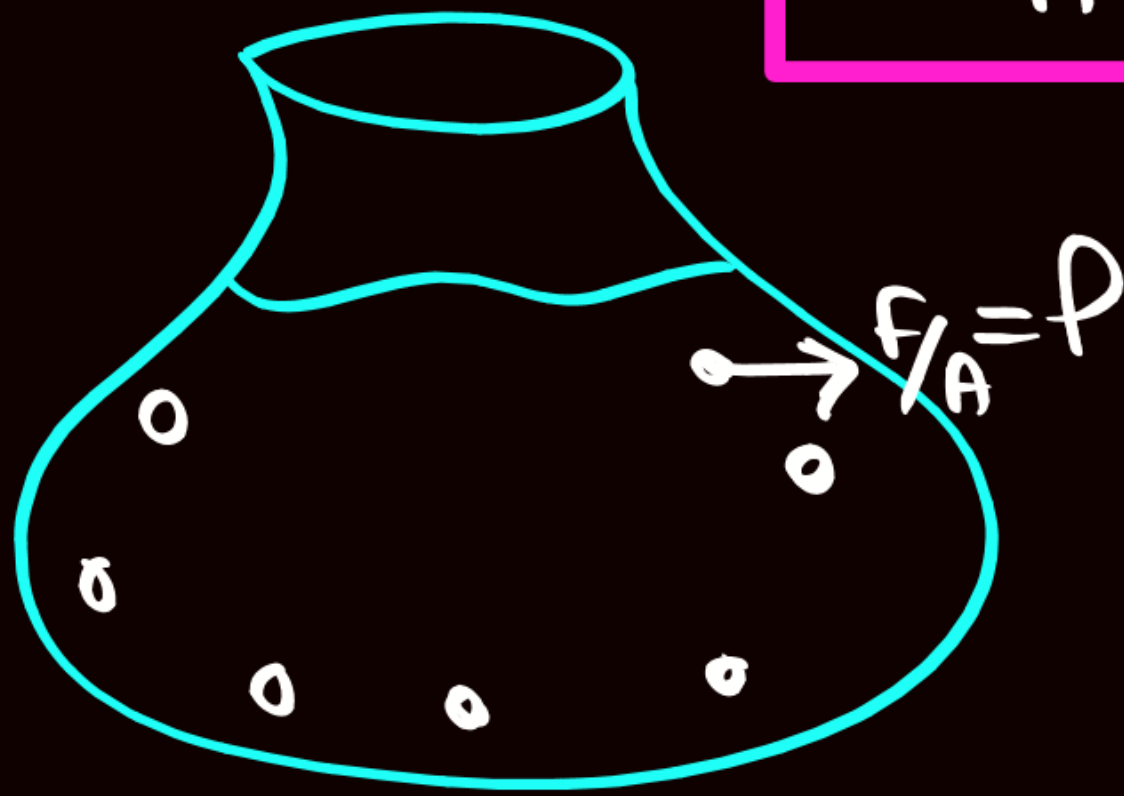
$$\downarrow \rho = \frac{m \downarrow}{V \uparrow}$$

$$p_1 > p_2$$

Pressure

- It is the force (F)
Acting per unit
Area (A)

$$P = \frac{F}{A}$$



unit of pressure

4

$$P = \frac{F}{A} = \frac{N}{m^2} = \text{Pascal (S.I)}$$

Other units of pressure

atm

Bar

Cm of Hg

mm of Hg

torr



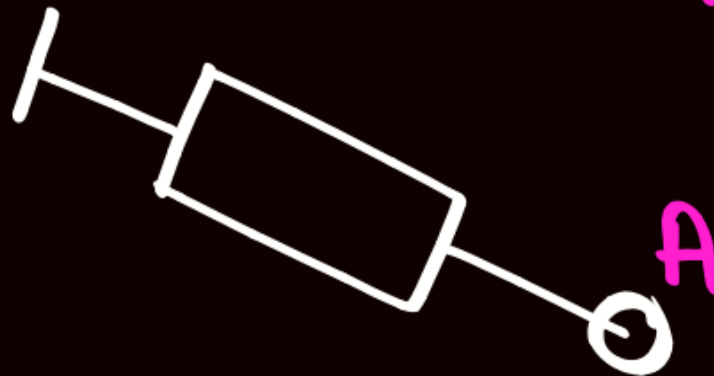
Application of Pressure 5

(1) Foot of Elephant



$$\downarrow\downarrow P = \frac{F}{A} \uparrow\uparrow\uparrow$$

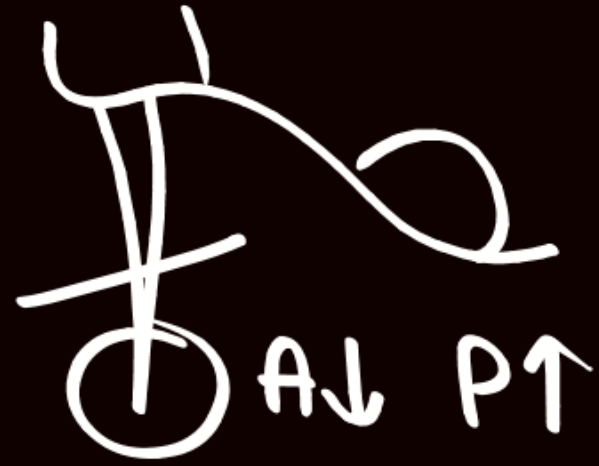
(2) Injection



$$\uparrow\uparrow\uparrow P = \frac{F}{A} \downarrow\downarrow$$

(3) Ladies heel

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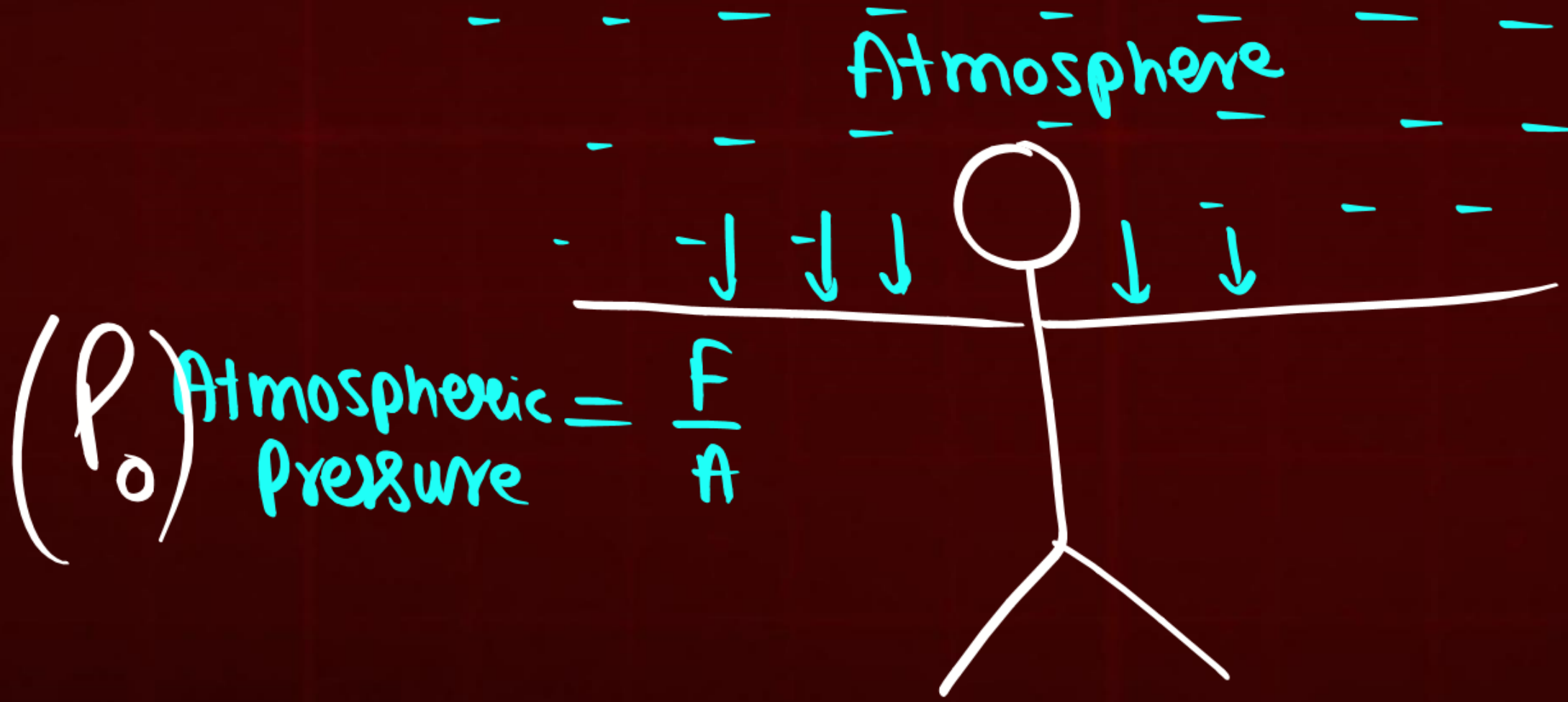
(4) Boys



Atmospheric Pressure



The force exerted by the atmosphere per unit area is called Atmospheric pressure

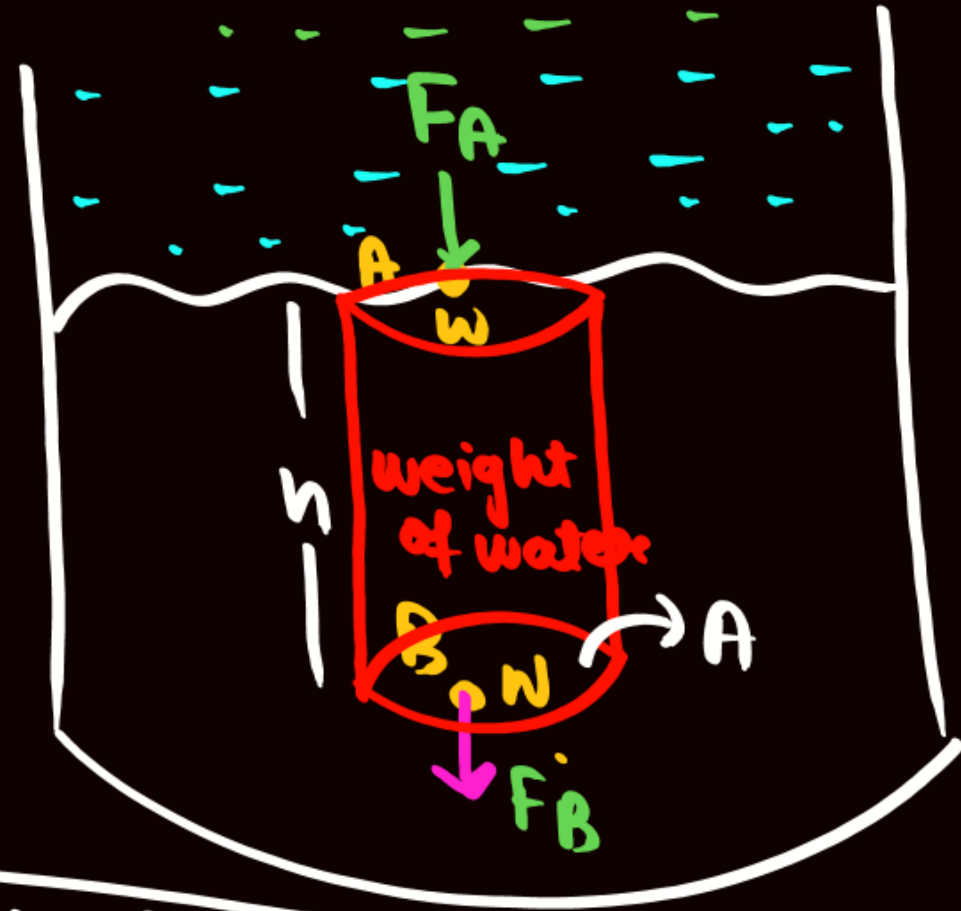


Variation of pressure with depth

⑧

- As we move down the liquid the pressure ↑↑↑↑

Atmosphere



qth

$$V = A \times h$$

Math

$$F_B = F_A + \text{weight of liquid.}$$

$$F_B = F_A + W$$

$$F_B = F_A + \textcircled{mg}$$

$$F_B = F_A + V \rho g$$

$$F_B = F_A + A h \rho g$$

$$\frac{F_B}{A} = \frac{F_A}{A} + \frac{A h \rho g}{A}$$

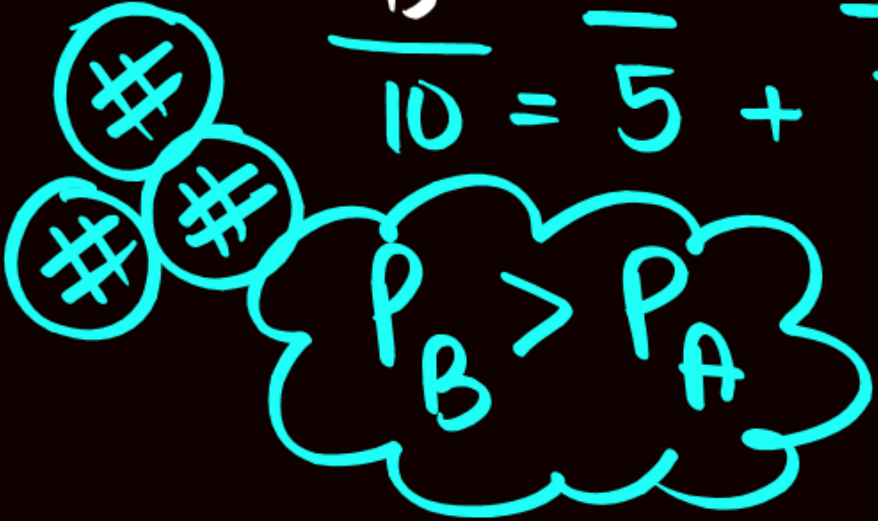
$$P_B = P_A + h \rho g$$

$$\rho = \frac{m}{V}$$

$$V \rho = m$$

$$P_B = P_A + \rho g h$$

$$\frac{10}{5} = \frac{5}{5} + \frac{5}{5}$$



$\rho \rightarrow$ Density of liquid (9)

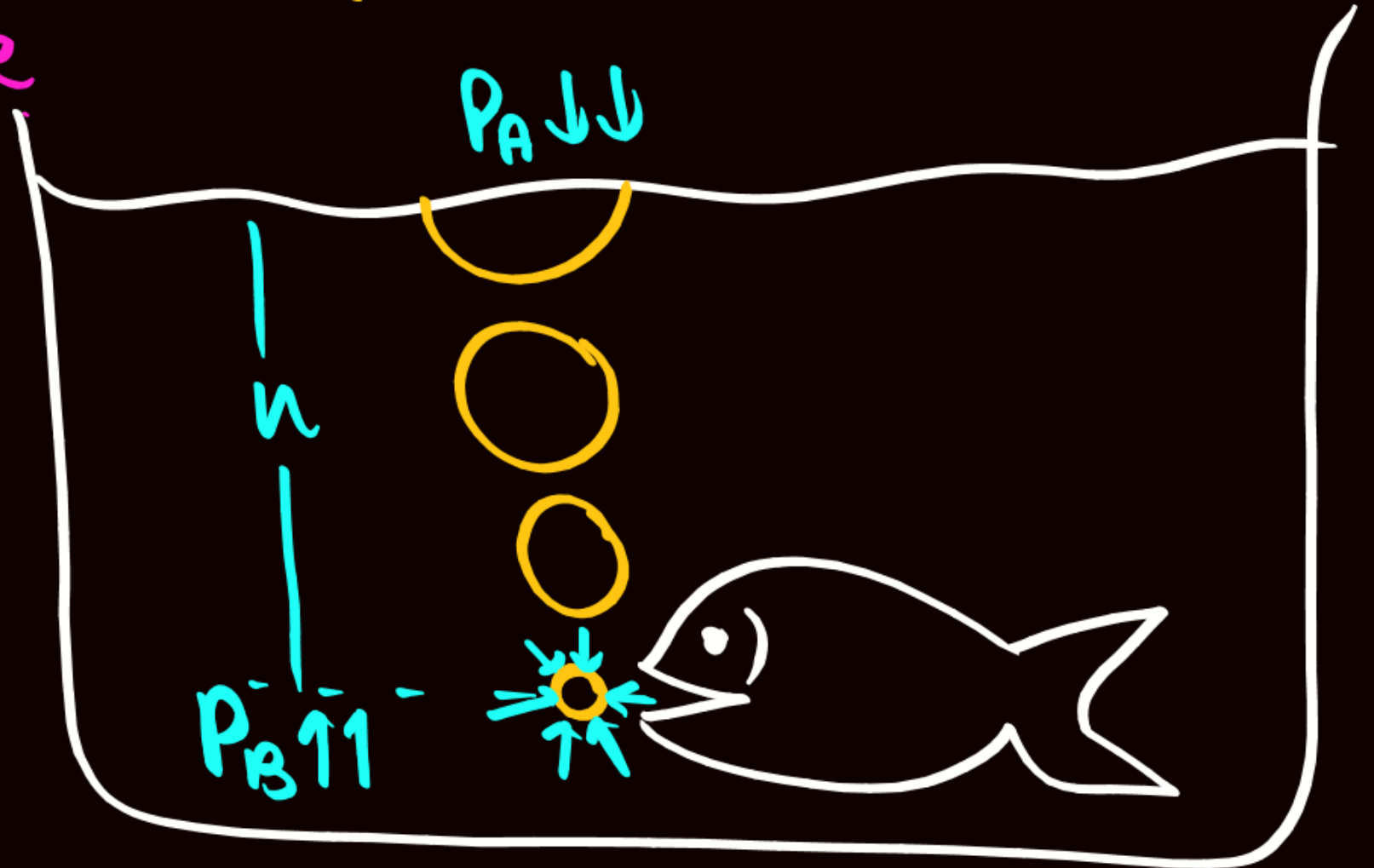
$g = 10 \text{ m/s}^2$ (all due to gravity)

Hydrostatic pressure $h \Rightarrow$ height in the liquid

$$P_B = P_A + \rho g h$$

Absolute
Pressure
at B

Atmospheric
Pressure (P_0)



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$$p_B = \cancel{p_A}^{p_0} + \rho gh$$

$$p_B = p_0 + \rho gh$$

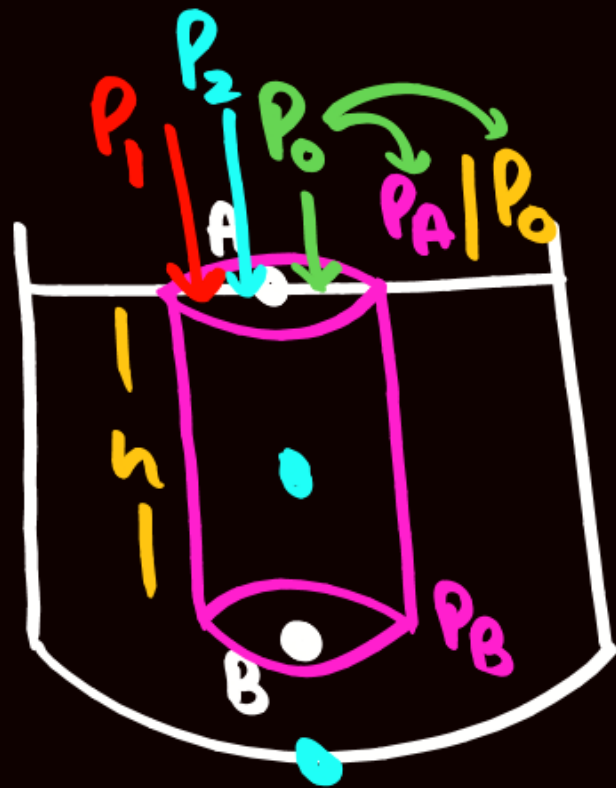
Pascal law

Summer - 2026

3M

11

- A pressure applied to any point of a liquid, is transmitted equally to every portion of the liquid



$$\# P_B = P_A + \rho gh$$

$$\# P_B = P_1 + P_A + \rho gh$$

$$\# P_B = P_1 + P_2 + P_A + \rho gh$$

Application of pascal law

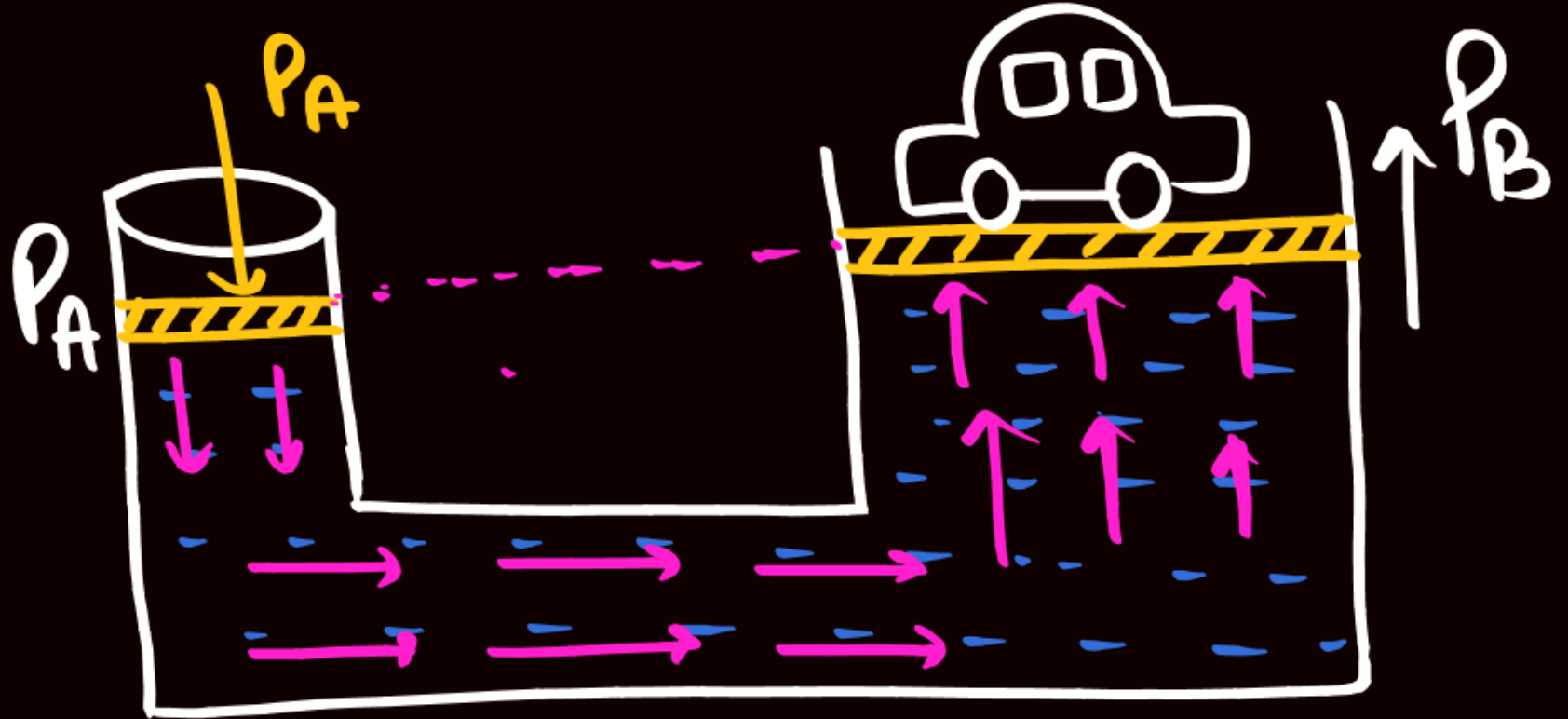
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① Hydraulic lift.

qth

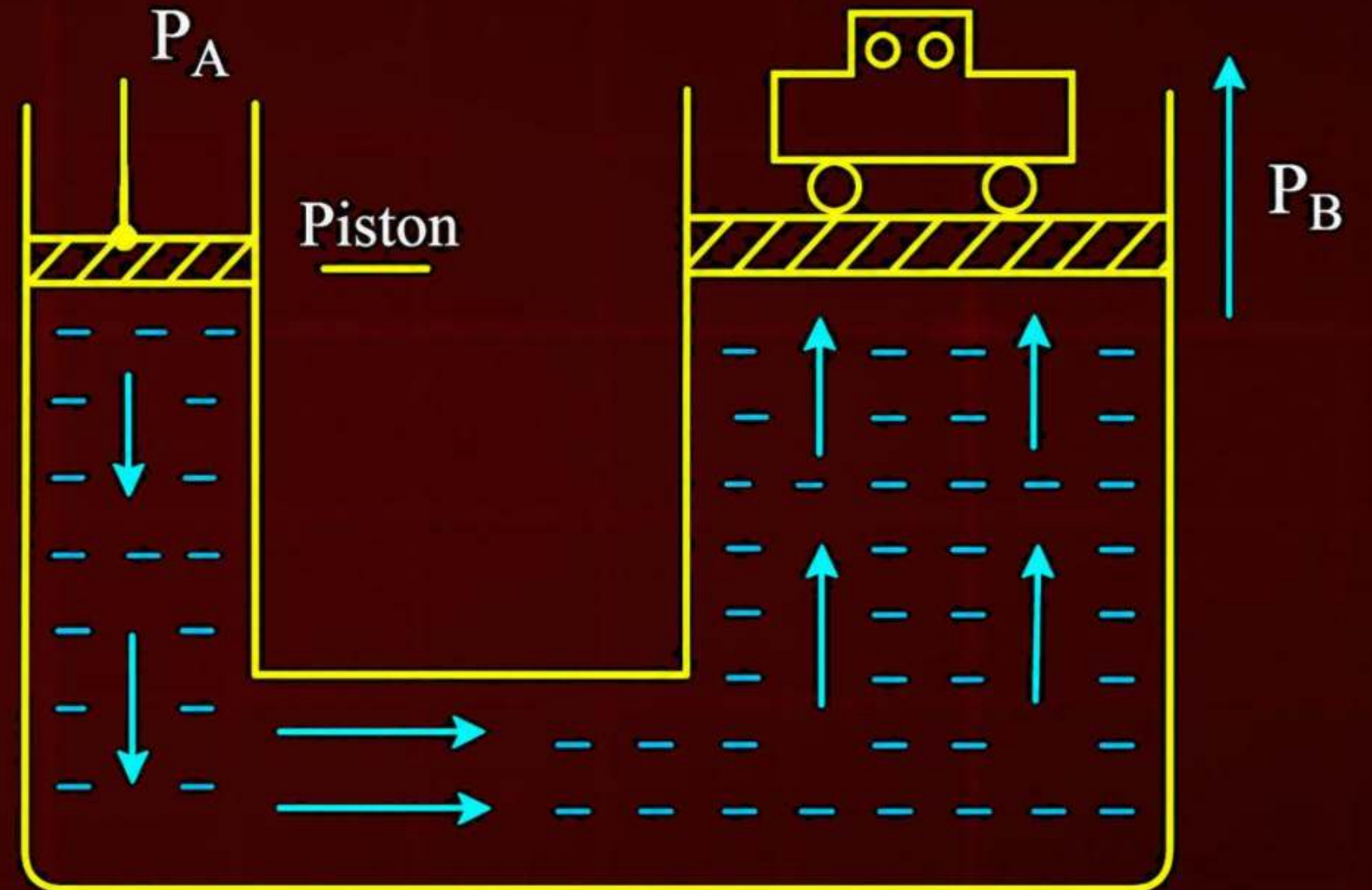
$$P_A = P_B$$

Piston



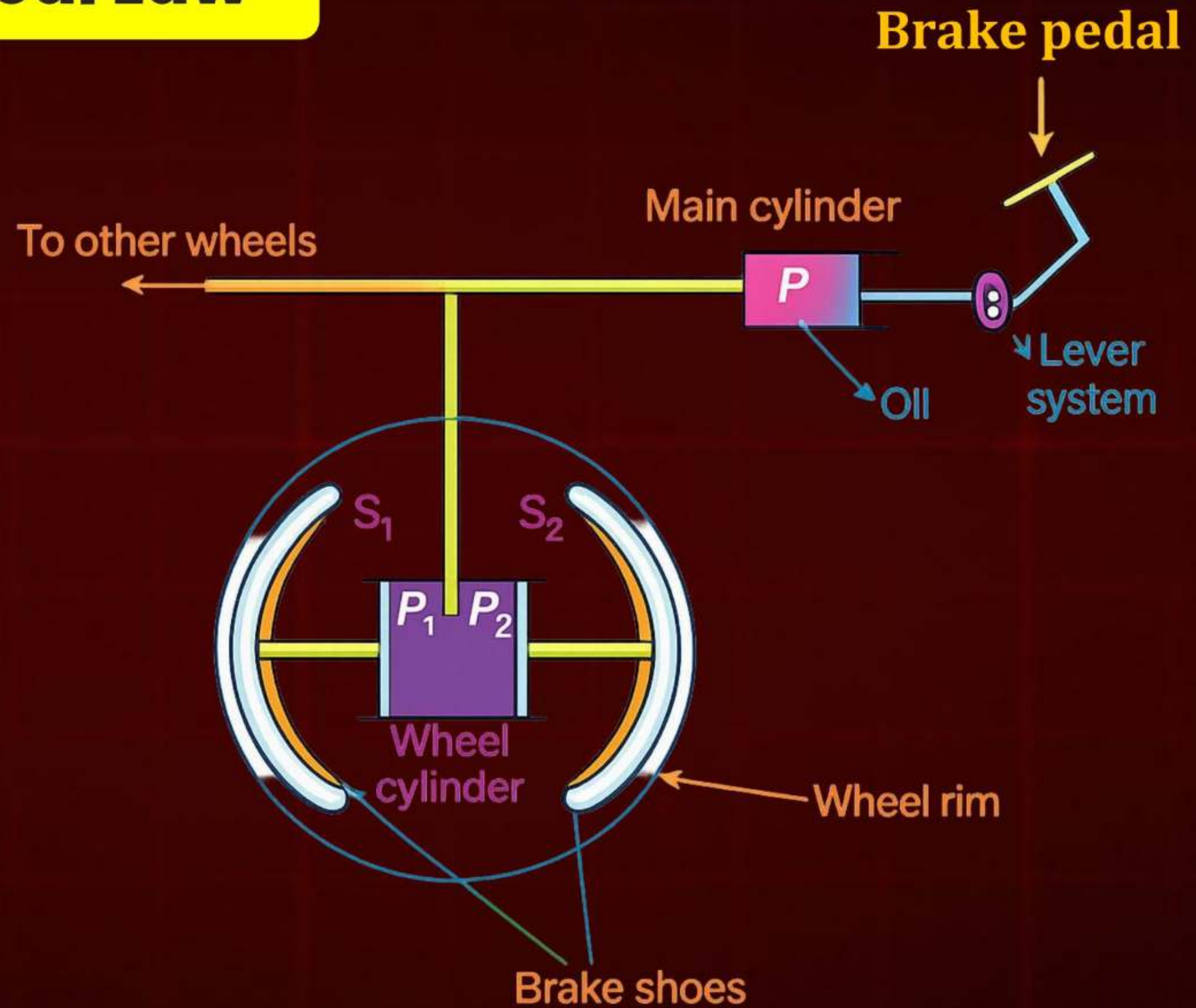
Application of Pascal Law

1. Hydraulic Lift



Application of Pascal Law

2. Hydraulic Brakes



Streams And Turbulent Flow

Stream line flow

It is the type of flow in which liquid particle follow the fixed path.

Turbulent Flow

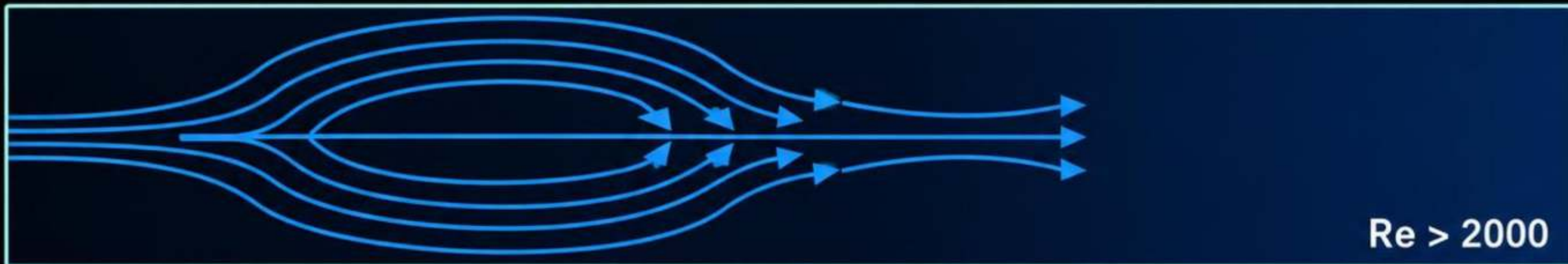
It is the type of flow in which liquid particle does not follow the fixed Path.

Reynolds Number (Re)

Laminar Flow



Turbulent Flow



$$Re = \frac{\rho V L}{\mu}$$

ρ : Density of the fluid

V : Velocity of flow

μ : Viscosity of the fluid

L : Characterizing length of the flow system
(usually the diameter of the pipe)

Equation of Continuity

The volume of liquid in is equal volume of liquid out

HOMework

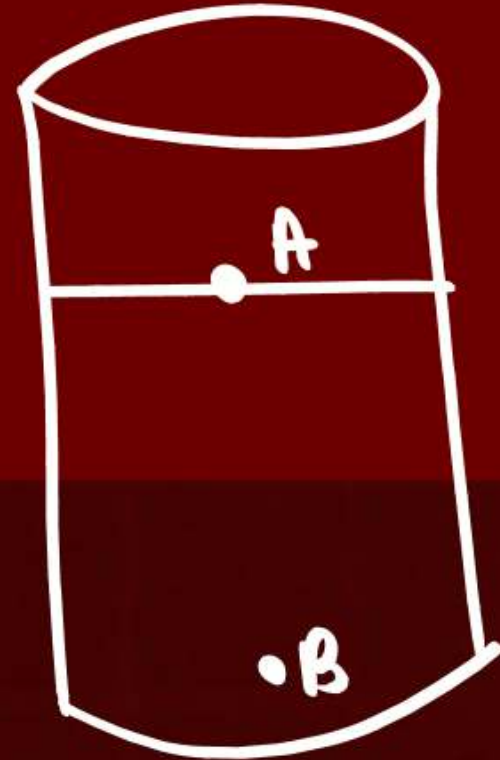
Q Numerical

(a) $P_A = P_B$

(b) $P_B < P_A$

(c) $P_B > P_A$

(d) Main gada (Dockey) nu.



“Jab Ek Mard Padhhta hai, Toh Sirf
ek Aadmi Padhhta hai. Lekin Jab ek
Aurat Padh Jaati hai, Toh Poora
Khandaan Padh Jata hai.”



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